

Server-based vs Hypervisor-based virtualization

- In server-based virtualization, multiple applications or services run on a single server's operating system. However, each service operates in an isolated environment.
- This form of virtualization is commonly done through containers or virtual environments like Docker or Kubernetes.
- Hypervisor-based virtualization involves creating multiple virtual machines (VMs), each with its own operating system, on a single physical server.
- This setup uses a hypervisor (e.g., VMware ESXi, Microsoft Hyper-V) to manage and isolate each VM, allowing them to run independently.



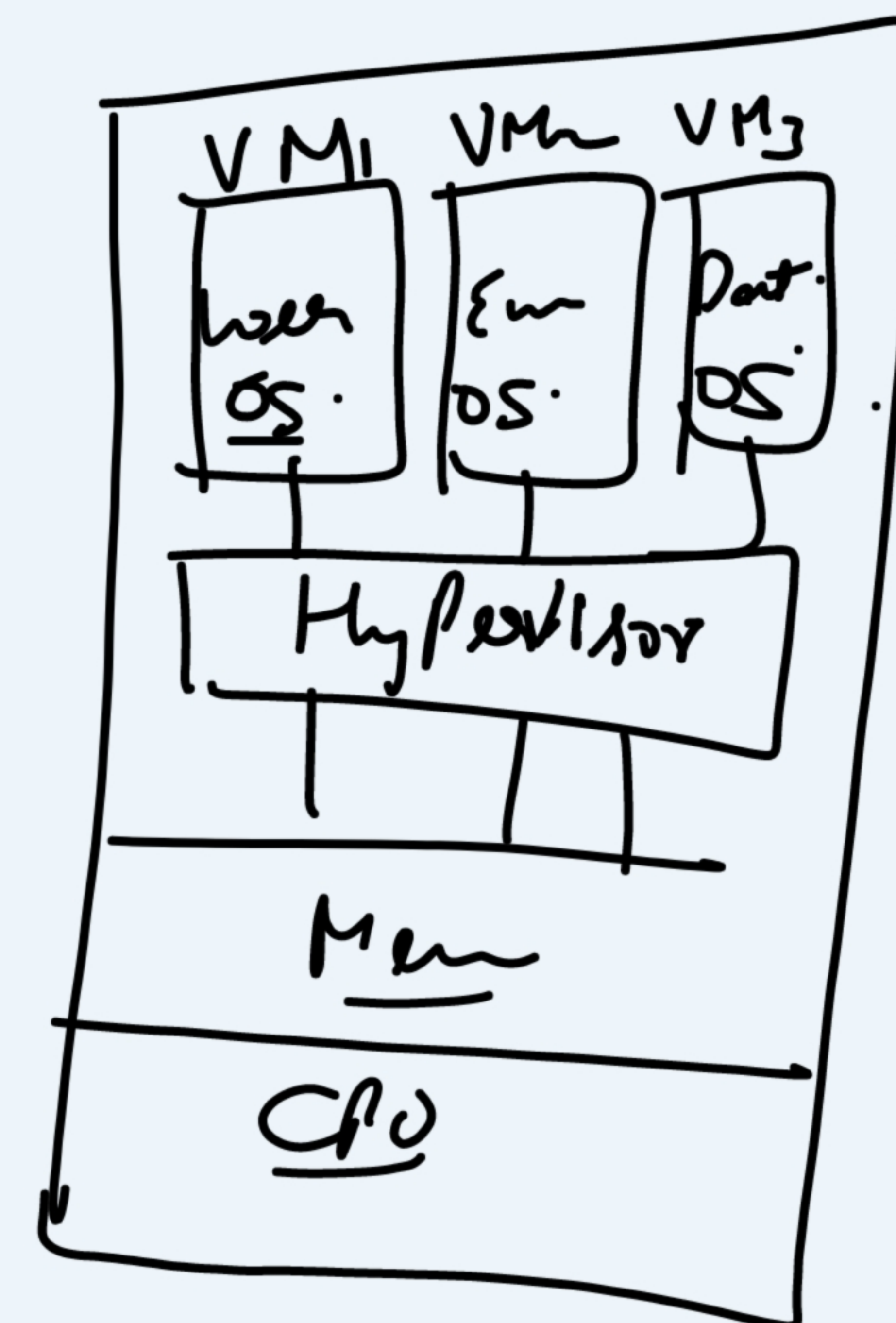
Server

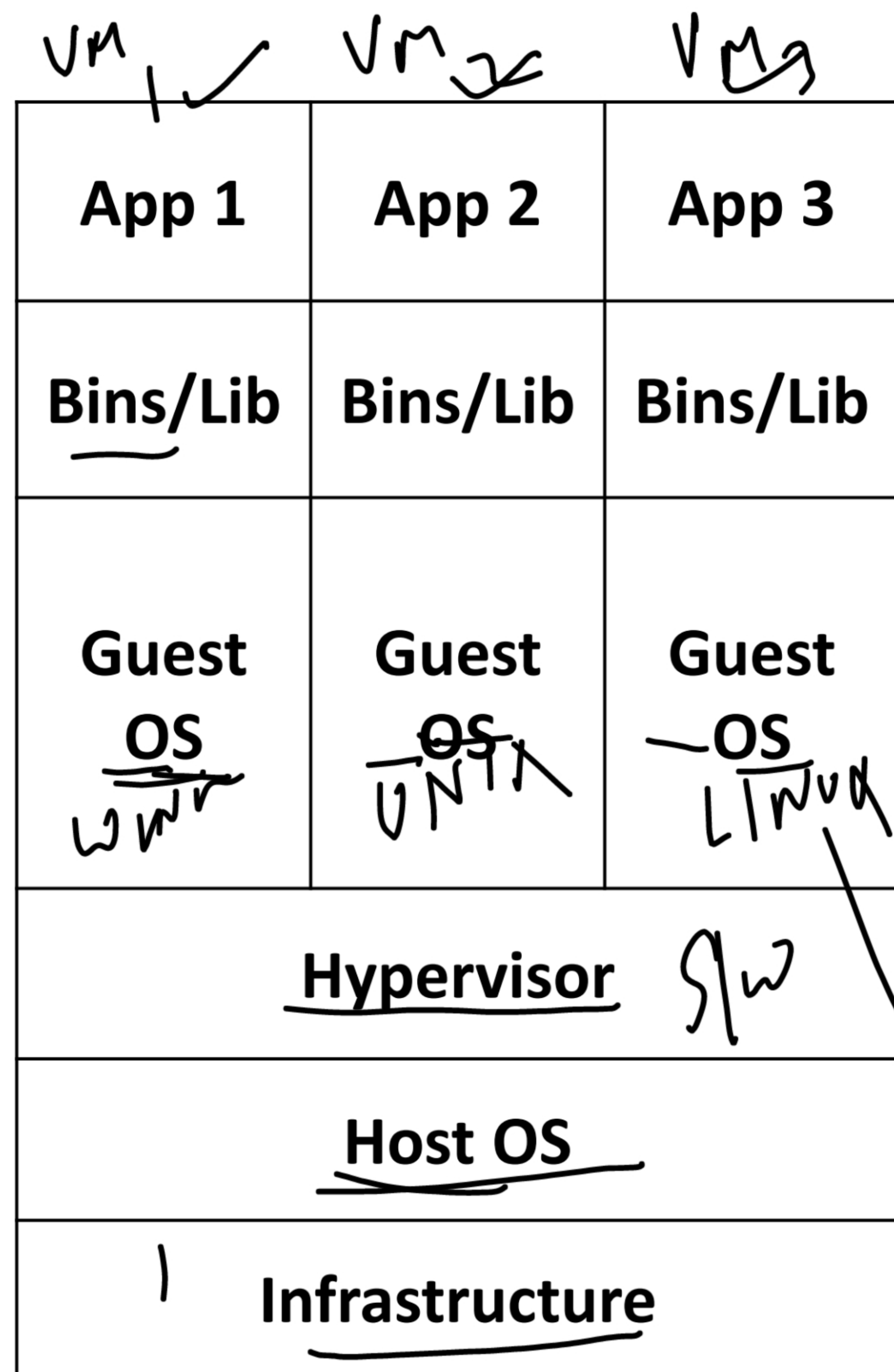


Server



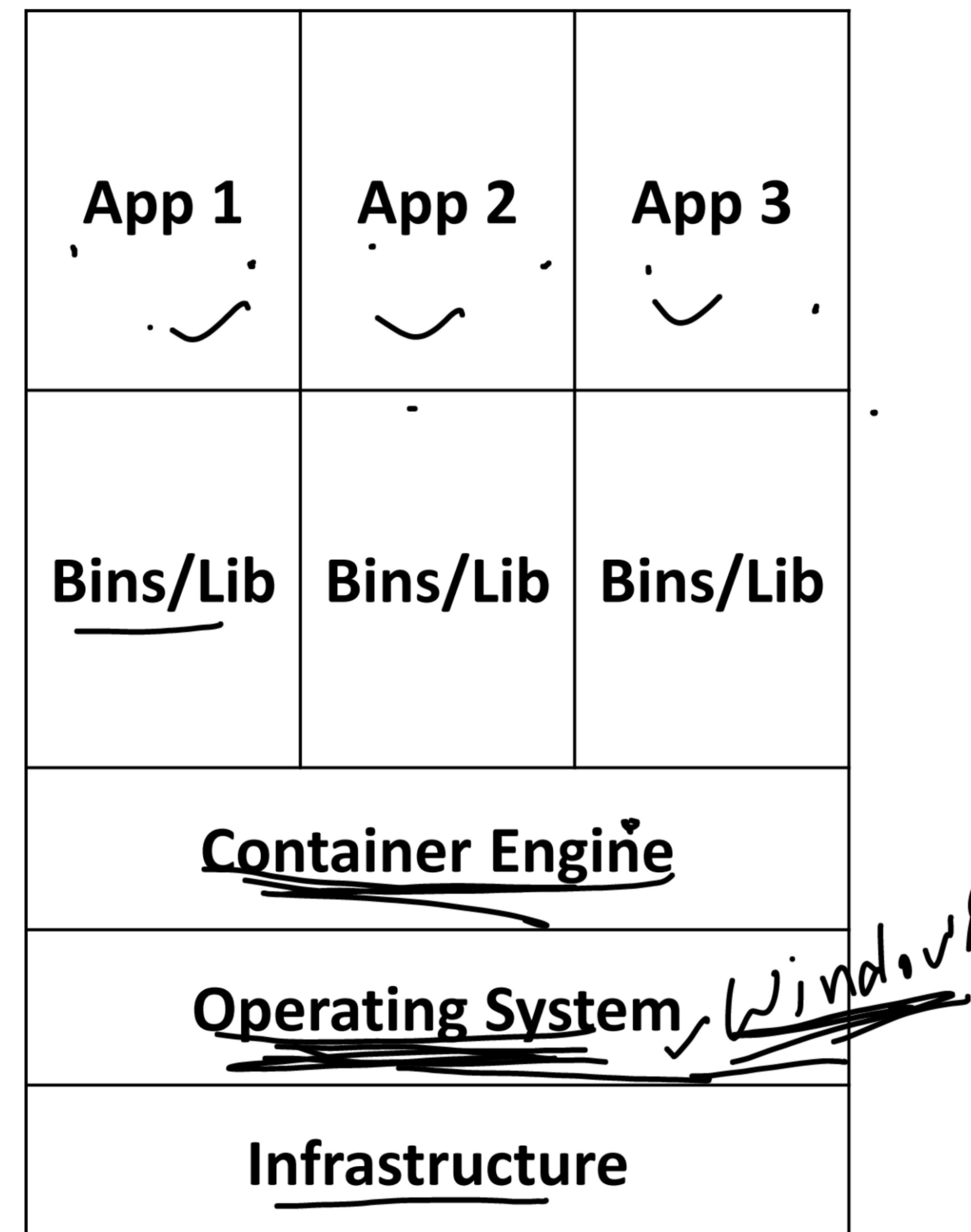
Server





Virtualization

9-2-1



Containerization

Containers	VMs
Containers virtualize at the <u>application layer</u>	VMs virtualize at the <u>hardware level</u> , VMs run on a <u>hypervisor</u>
Containers are lightweight ✓	VMs are heavier ✓
Containers are faster to start ✓	VMs take longer to start ✓
More efficient in <u>terms of memory and CPU usage</u>	Less Efficient Comparatively ✓
<u>Containers have less isolation than VMs.</u>	<u>More Isolation that's why VMs more secure</u>
Containers are highly portable	VMs are less portable
Ideal for <u>microservices architecture</u> and <u>stateless applications</u>	Ideal for monolithic applications and legacy systems
Netflix (microservices), Spotify (CI/CD), Shopify (scalability), GitLab (development), AWS Lambda (serverless functions)	JP Morgan Chase (financial security), Cerner (healthcare compliance), Pentagon (government security), Microsoft Azure (multi-OS), NASA (scientific research)